Chapter 12: The Van der Pol Oscillator

(Dated: May 04, 2020)

BACKGROUND

- Why this area of classical mechanics is important:

The chapter I was given for the final project was chapter 12. Chapter 12 is about nonlinear mechanics and chaos. The topic of my project is the Van der Pol oscillator. In dynamics, the Van der Pol oscillator is a non-conservative oscillator with non-linear damping. The Van der Pol oscillator was created by a Dutch electrical engineer and physicist Balthasar van der Pol. He found stable oscillations which at the time he had called relaxation oscillations, which are now known as a type of limit cycle in electrical circuits. When these circuits are driven near the limit cycle, they become entrained. The Van der Pol equation springs up in a wide variety of physics and engineering problems, as it describes an oscillator with some non-linear spring damping. In particular, variations on the equation can describe complex circuit interactions with vacuum tubes, action potentials for neurons, or seismology models of plate interaction at a fault line.

- Core concepts that need to be understood for the solution to make sense:

Limit Cycle: Initially, the orbits in phase space vary with the difference in the initial values; they later merge into a single circuit and apparently rotate along the circuit. Thus, the final circuit does not depend on the initial conditions. This type of orbit is called a limit cycle. In this problem I investigate different limit cycles, so having a understanding of what it is, is important.

Harmonic oscillator: In mechanics, a harmonic oscillator is a system, that when displaced from its equilibrium position, experiences a restoring force F proportional to the displacement. The van der Pol oscillator is a form of a harmonic oscillator with an angular frequency of unity.

Chaos: If the drive strength is increased beyond certain parameters and initial conditions, the long-term motion becomes nonperiodic, and we say that chaos has set in. Chaotic motion is extremely sensitive to initial conditions.

- Key Equations:

Van der Pol equation:

Driven System: $\ddot{x} - \varepsilon(1 - x^2)\dot{x} + \alpha x = Fcos(\frac{2\pi t}{T})$

Non-Driven System: $\ddot{x} - \varepsilon(1 - x^2)\dot{x} + \alpha x = 0$

As stated above, the van der Pol equation comes from the basics equation of a harmonic oscillator, where the $\varepsilon(1 - x^2)\dot{x}$ term is the non-linear spring damping.

- Explanations of any tricky concepts that caused me to stumble along the way:

One of the definitions that I stumbled on was a phase space diagram. This is just a plot of the velocity vs the position. Position going on the x-axis and velocity on the y-axis.

Another concept I stumbled on was the idea of a limit cycle. But after doing some research, it is easy to see that it is just an orbit that can be stable or unstable.

Another concept was the idea of rationalizing whether a point or the limit cycle was stable or unstable. But in order to decide if something is stable graphically, it is only a matter of seeing if you start near the equilibrium point, do you stay near the point or away from the point. If you stay near the point then you have a case of stability and if you go away from it you have a case on unstability.

THE PROBLEM

Task (A), Non-Driven System: Initially we'll look at the non-driven system, so let $F = 0$

- Part (i) Determine the location of any equilibrium points

Initial equation: $\ddot{x} - \varepsilon(1 - x^2)\dot{x} + \alpha x = 0$

Convert it to the first order ODE system. Let: $y_1 = x$, $y_2 = \dot{x}$

Then substitute y_1, y_2 into the initial equation and obtain: $\begin{cases} \dot{y}_1 = y_2 \\ \dot{y}_2 = \varepsilon(1 - y_1^2)y_2 - \alpha y_1 \end{cases}$

To define the location of the equilibrium point one should equal right hand sides of these equations to zero.

Thus, obtain: $\begin{cases} y_2 = 0 \\ \varepsilon(1 - y_1^2)y_2 - \alpha y_1 = 0 \end{cases}$

The only possible solution is: $y_1 = y_2 = 0$.

-Part (ii) To see if these points are stable or unstable, numerically solve and plot a phase space diagram of the solution for an object that initially had $x_0 = 0.1$ and $\dot{x_0} = 0$. Do your equilibrium points look like they'd be stable or unstable.?

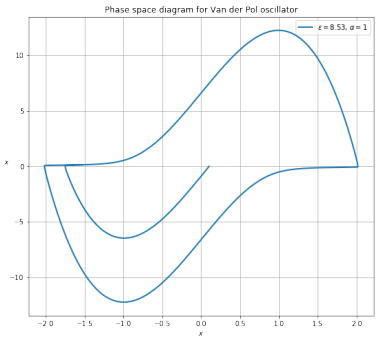

FIG. 1: Phase Space Diagram for Van der Pol Oscillator

According to this diagram above, the system being released at the neighborhood of equilibrium point goes away from it. Thus, I can conclude, that point $y_1 = y_2 = 0$ is unstable equilibrium.

-Part (iii) Plot a quiver plot overtop of your plotted solution above. This will help you see the general movement and path that an object would follow for a variety of initial conditions.

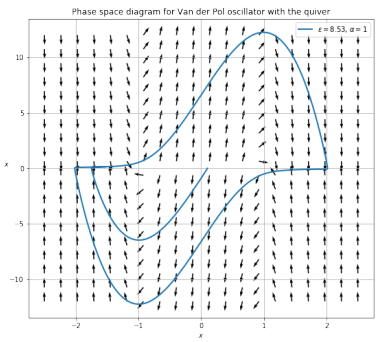

FIG. 2: Quiver Plot for Van der Pol Oscillator

You can see from the quiver plot, that the general direction in which follows the phase diagram.

-Part (iv) Matplotlib offers a nicer way to see this behavior with a streamplot. Create one covering the same region as your above plot to see how they compare. (This is actually a case where the streamline does not capture the motion as nicely.)

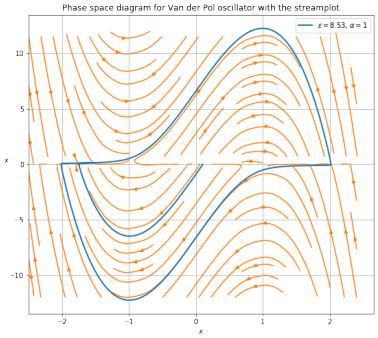

FIG. 3: Streamplot for Van der Pol Oscillator

-Part (v) While you may not have many equilibrium points, your plots should suggest the existence of something new: an equilibrium orbit, also called a limit cycle. Is this orbit stable or unstable?

According to the previous plot, the limit cycle seems to be stable. Because streamlines from any point of phase diagram are pointed toward this cycle. Thus, from any initial conditions the system will tend to go to the equilibrium orbit.

-Part (vi): You can get the shape of just the orbit by numerically solving the equation over many cycles and then only plotting the later cycles. Do this to investigate the shape of the limit cycle when $\epsilon = 0, 1, 4, 8.53$.

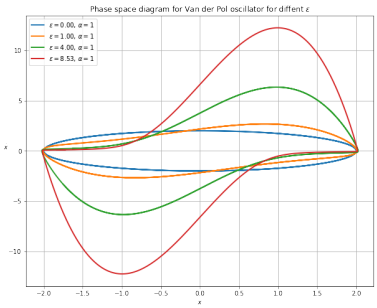

FIG. 4: Limit Cycles for Van der Pol Oscillator

As you increases epsilon, you can see that the amplitude increases.

Task (B), Driven System: Now we'll look at the driving system. Let's set $F = 0$ and $T = 20$.

-Part (i): Solve the differential equation numerically for two different initial conditions, one with $x_1 = 2, \dot{x_1}1 = 0$ and the other with $x_2 = 2.05, \dot{x_2} = 0$. Plot the two solutions versus time over about 20 seconds. The two solutions started very close together. How far apart are they after 30 seconds?

Initial equation: $\ddot{x} - \varepsilon(1 - x^2)\dot{x} + \alpha x = F\cos\left(\frac{2\pi t}{T}\right)$
Convert it to the first order ODE system. Let: $y_1 = x$, $y_2 = \dot{x}$
Then substitute y_1, y_2 into the initial equation and obtain: $\begin{cases} \dot{y}_1 = y_2 \\ \dot{y}_2 = F\cos\left(\frac{2\pi t}{T}\right) + \varepsilon(1 - y_1^2)y_2 - \alpha y_1 \end{cases}$

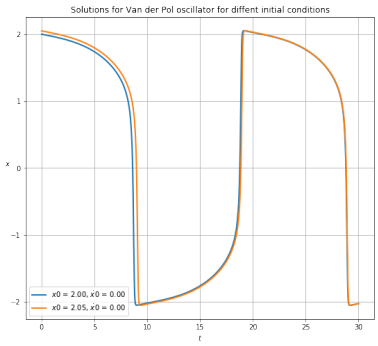

FIG. 5: Two Solutions vs Time Graph

-Part (ii): Plot the phase space of both solutions. Does this motion seem chaotic?

As shown from the graph below, even with a force driving the oscillator, the motion does not seem to be chaotic. There is various initial conditions that one could put and it would still lead to nice stable limit cycles even with forced oscillations.

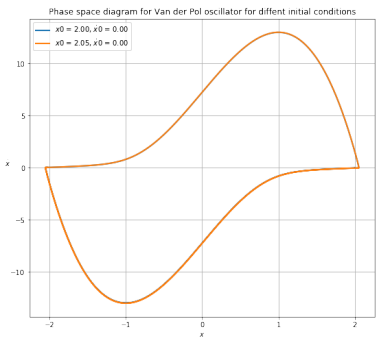

FIG. 6: Phase Space Diagram 2 for Van der Pol Oscillator

Task (C), Driven system, $\omega = 1.788$. Repeat part (B) above, but now with $\omega = 1.788$. Also using parameters of $\epsilon = 3$, $F = 5$, and $a = 1$,

-Part (i): Solution vs Time

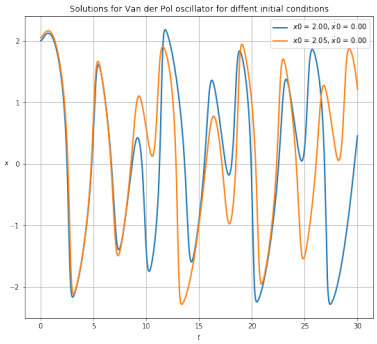

FIG. 7: Solution vs Time 2 for Van der Pol Oscillator

-Part (ii): Phase Diagram

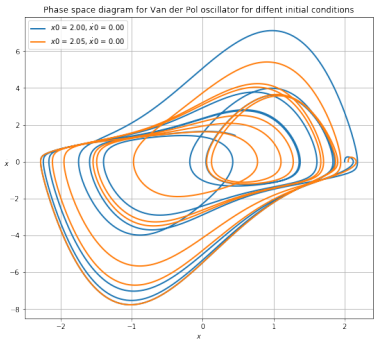

FIG. 8: Phase Space Diagram for Van der Pol Oscillator

After looking at the above figures (7) and (8), it is apparent that when you add the angular frequency of about 1.788, and change the parameter of epsilon, F, and a, you can see that this leads to a more chaotic motion.

References

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Kinoshita, Shuichi. "Van Der Pol Oscillator." *Van Der Pol Oscillator - an Overview | ScienceDirect Topics*, 2013, www.sciencedirect.com/topics/engineering/van-der-pol-oscillator.

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